Lab 9: Inference for Difference of Two Means and Paired Data

DATASCI 100: Introduction to Statistical Inference

Spring 2025

Overview

In previous labs we focused on making inferences about a single population mean. In this lab we extend those methods to settings where we want to compare means across two groups. We cover two closely related procedures.

In Part 1 we analyze **paired data**, where each observation in one group is naturally matched to an observation in the other group. We compute the difference within each pair and then run a one-sample t-test on those differences. In Part 2 we analyze **two independent samples**, where the two groups have no natural pairing, and we run a two-sample t-test to compare their means directly.

Both parts use the tidyverse for data import, wrangling, and visualization.

Data

Part 1 – Course Evaluations

This data set contains course evaluation information for 463 classes taught by 94 randomly selected professors at the University of Texas at Austin. The data are in `CourseEvals.csv`. We focus on two variables: `course_eval`, the average course evaluation score (1–5), and `prof_eval`, the average professor evaluation score (1–5). Because both scores are recorded for the same course, they are naturally paired.

Note: Because one professor may teach multiple courses, the observations are not truly independent. For simplicity in this course, please treat observations as independent.

Part 2 – Mario Kart eBay Auctions

This data set records all eBay auctions for the Wii game *Mario Kart* during one week in October 2009. Auctions were included only if they met a set of conditions ensuring comparability (US-based, at least one bidder, selling the game itself, no Buy It Now completion, etc.). The data are in `mariokart.csv` and contain 12 variables; the ones relevant to this lab are described in Table 1.

Table 1: Selected variables in `mariokart.csv`

Variable	Description
`total_pr`	Total price (auction price + shipping price, in USD)
`ship_sp`	Shipping speed or method
`cond`	Game condition: new or used
`n_bids`	Number of bids
`wheels`	Number of Wii wheels included

Part 1: Paired t-Test

Loading the Data

```r
library(tidyverse)

# Import course evaluations
evals <- read_csv("CourseEvals.csv")
```

Motivation

In Lab 8 we tested whether the average percent of students completing evaluations differed from 80%. Now we ask a different question: are professor evaluation scores significantly different from course evaluation scores? These two variables are recorded for the same course and are therefore *not* independent of one another—a professor who receives a high personal score likely also receives a relatively high course score. When two measurements come from the same unit of observation, we should use a **paired t-test** rather than a two-sample t-test.

Other common paired settings include pre- and post-treatment scores (e.g., does blood pressure change after an intervention?), or measurements taken on the same subject under two different conditions.

Computing and Visualizing the Differences

The paired t-test reduces the two-variable problem to a one-variable problem by computing the within-pair difference for each observation. We then run a one-sample t-test on those differences.

```r
# Compute the difference for each course
evals <- evals %>%
  mutate(diff = course_eval - prof_eval)

# Summary statistics for the difference
evals %>%
  summarize(
    mean_diff = mean(diff),
    sd_diff   = sd(diff)
  )

# Visualize the distribution of differences
ggplot(evals, aes(x = diff)) +
  geom_histogram(bins = 20) +
  labs(x = "course_eval - prof_eval",
       title = "Distribution of Paired Differences")
```

The average difference is approximately -0.18. The negative sign tells us that course evaluations tend to be *lower* than professor evaluations. The histogram shows that the differences are approximately normally distributed, which satisfies the normality condition for the paired t-test.

Conducting the Paired t-Test

Our hypotheses are

$$H_0 : \mu_{\text{diff}} = 0 \quad \text{vs.} \quad H_a : \mu_{\text{diff}} \neq 0,$$

where $\mu_{\text{diff}} = \mu_{\text{course}} - \mu_{\text{prof}}$ is the true average within-pair difference. We can carry out this test in two equivalent ways.

```
# Method 1: one-sample t-test on the difference variable
t.test(evals$diff)

# Method 2: supply both variables directly and set paired = TRUE
t.test(evals$course_eval, evals$prof_eval, paired = TRUE)
```

Both methods produce identical output. Method 1 makes the logic transparent because we can see the distribution of differences before testing. Method 2 is more convenient when you have not already created a difference column; R computes the differences internally by subtracting the second variable from the first.

The test statistic is $t = -19.16$ with 462 degrees of freedom. At $\alpha = 0.05$ we reject H_0 ($p < 0.001$) and conclude that the average difference is significantly different from zero. We are 95% confident that the true average difference lies between -0.19 and -0.16. Because the entire interval is negative ($\mu_{\text{course}} - \mu_{\text{prof}} < 0$, i.e., $\mu_{\text{course}} < \mu_{\text{prof}}$), there is strong evidence that professors receive higher personal evaluations than their courses do.

Part 2: Two-Sample t-Test

Loading and Cleaning the Data

```
# Import Mario Kart data
mariokart <- read_csv("mariokart.csv")

# Quick overview
glimpse(mariokart)
summary(mariokart$total_pr)
```

Exploring the Outcome Variable

We are interested in `total_pr`, the final total price of each auction. Before comparing groups, we inspect the overall distribution.

```
ggplot(mariokart, aes(x = total_pr)) +
  geom_histogram(bins = 20) +
  labs(x = "Total Price (USD)", title = "Distribution of Total Price")
```

The histogram reveals a small number of auctions with unusually high prices. Examining these outliers shows that they included multiple games—they are not comparable to a single copy of Mario Kart.

```
# Identify high-price outliers
mariokart %>%
  filter(total_pr > 100)


# We remove these observations to obtain a clean analysis data set.


```

```r
# Remove outliers
mkClean <- mariokart %>%
  filter(total_pr < 100)

# Verify the distribution after cleaning
ggplot(mkClean, aes(x = total_pr)) +
  geom_histogram(bins = 20) +
  labs(x = "Total Price (USD)", title = "Total Price After Removing
     Outliers")
```

Use `mkClean` for all remaining work in Part 2.

Recoding the Shipping Speed Variable

The variable `ship_sp` has several categories, some with very few observations. We collapse them into four groups for a cleaner analysis.

```r
# Recode shipping speed into four groups
mkClean <- mkClean %>%
  mutate(newship = case_when(
    ship_sp %in% c("firstClass", "priority") ~ "FirstClass/Priority",
    ship_sp %in% c("ups3Day", "upsGround")   ~ "UPS",
    ship_sp == "standard"                    ~ "Standard",
    TRUE                                     ~ "other"
  ))

# Verify the recoding
table(mkClean$newship, mkClean$ship_sp)
```

The `case_when()` function is the tidyverse equivalent of a series of nested `if/else` statements. Each line specifies a condition and the value to assign when that condition is true; the final `TRUE` acts as a catch-all for any remaining rows.

Motivation for the Two-Sample t-Test

Suppose we want to know whether UPS-shipped auctions sell for a different average price than Standard-shipped auctions. These two groups contain *different* auctions with no natural pairing between them, so a paired test is not appropriate. Instead we use a **two-sample t-test**, which directly compares the means of two independent groups.

Our hypotheses are

$$H_0 : \mu_{\text{UPS}} = \mu_{\text{Standard}} \quad \text{vs.} \quad H_a : \mu_{\text{UPS}} \neq \mu_{\text{Standard}},$$

where μ_{UPS} and μ_{Standard} are the true average total prices for each shipping method.

Exploring the Two Groups

Before testing, we examine each group's distribution.

```r
# Filter to the two groups of interest
mk_two <- mkClean %>%
  filter(newship %in% c("UPS", "Standard"))
```

```r

# Side-by-side boxplots
ggplot(mk_two, aes(x = newship, y = total_pr)) +
  geom_boxplot() +
  labs(x = "Shipping Method", y = "Total Price (USD)",
       title = "Total Price by Shipping Method")

# Histograms for each group
ggplot(mk_two, aes(x = total_pr)) +
  geom_histogram(bins = 15) +
  facet_wrap(~ newship) +
  labs(x = "Total Price (USD)")

# Group means and standard deviations
mk_two %>%
  group_by(newship) %>%
  summarize(n    = n(),
            mean = mean(total_pr),
            sd   = sd(total_pr))
```

Conducting the Two-Sample t-Test

```r
# Extract price vectors for each group
ups      <- mkClean$total_pr[mkClean$newship == "UPS"]
standard <- mkClean$total_pr[mkClean$newship == "Standard"]

# Two-sample t-test assuming equal variances
t.test(ups, standard, var.equal = TRUE)
```

The `var.equal = TRUE` argument tells R to pool the two sample variances when estimating the standard error, which is appropriate when we have reason to believe the population variances are similar. To relax this assumption and use the Welch approximation (which does not assume equal variances), set `var.equal = FALSE`—that is also the default behavior if the argument is omitted entirely.

The test statistic is $t = 2.50$ with 63 degrees of freedom. At $\alpha = 0.05$ we reject H_0 ($p = 0.02$) and conclude that the average total prices for UPS and Standard shipping differ significantly. We are 95% confident that the true difference in average prices ($\mu_{\text{UPS}} - \mu_{\text{Standard}}$) lies between \$1.21 and \$10.91. Because this interval lies entirely above zero, UPS auctions appear to sell for a higher average price than Standard-shipped auctions.

Choosing Between the Paired and Two-Sample t-Test

The key question when choosing between these two procedures is whether the observations in the two groups are naturally linked to each other. If each observation in one group is matched to a specific observation in the other—as with before-and-after measurements on the same subject—then the data are paired and you should use the paired t-test. If the two groups consist of entirely different and unrelated individuals or units, use the two-sample t-test. Applying the wrong test can lead to invalid inference.

Summary of Key Functions

Function	Purpose
`t.test(x)`	One-sample t-test (also used for paired differences)
`t.test(x, y, paired = TRUE)`	Paired t-test using two raw variables
`t.test(x, y, var.equal = TRUE)`	Two-sample t-test with pooled variance
`t.test(x, y, var.equal = FALSE)`	Two-sample t-test with Welch approximation (default)
`mutate(diff = x - y)`	Compute paired differences within a tibble
`filter(group %in% c(...))`	Subset to selected groups
`case_when(...)`	Recode a variable using multiple conditions
`group_by() %>% summarize()`	Group-wise summary statistics
`facet_wrap(~group)`	Separate histogram panels by group

Ideas for this lab were obtained from lab materials written for OpenIntro Statistics by Andrew Bray and Mine Çetinkaya-Rundel, and from the OpenIntro *Mario Kart* case study.